

Code - Part

Q-1) The model failed on Lucknow because it was trained on Spacenet images that has mostly US and European buildings. Those buildings are large & well separated, thus their roofs are easily detected. In case of Lucknow buildings are smaller, denser and more irregular.

Q-2) My model detected 4638 buildings in the selected area. It missed the most building in dense residential areas and also near the roads and streets.

Q-3) Open buildings found more buildings because it was created using a much larger model trained on huge amount of satellite imagery. It has been optimised for building detection across india. A data scientist should use an existing dataset when it is already accurate & solves the problem efficiently instead of spending time building a new model.

★ Questions

- Q-1) Downloading imagery from server keeps the notebook small and allows ~~that~~ us to select any location without storing large image files.
- Q-2) Increasing zoom from 18 to 19 →
- More map tiles downloaded, thus time increased.
 - Image become more detailed
 - Still the small rooftops ~~is~~ remain undetected.
- Q-3) Lowering confidence value to 0.4 increases the number of detected buildings but also increases false detections.
Increasing it to 0.9 reduces false detections but may miss real building (In case of sandbox model the number of detected buildings = 0 for conf = 0.9)
- Q-5) Fine tuning starts from a model that has already learned useful image features. It only needs to adapt to the new dataset so it requires less data and training time.
- Q-6) mAP compares predictions with the ref. labels. If the labels are incomplete or not perfectly accurate, the score can be low even though many predicted buildings are correct.

Q-7) Building count is not enough to estimate population density as —

- different building ~~has~~ have different no. of floors
- buildings have different no. of people living in them.

Q-8) No, model training was not a waste of time. It actually helped to understand how object detection works.

But in practice existing datasets should be used when they are reliable. Custom models are useful where no suitable dataset already exist.